

Diagnosis

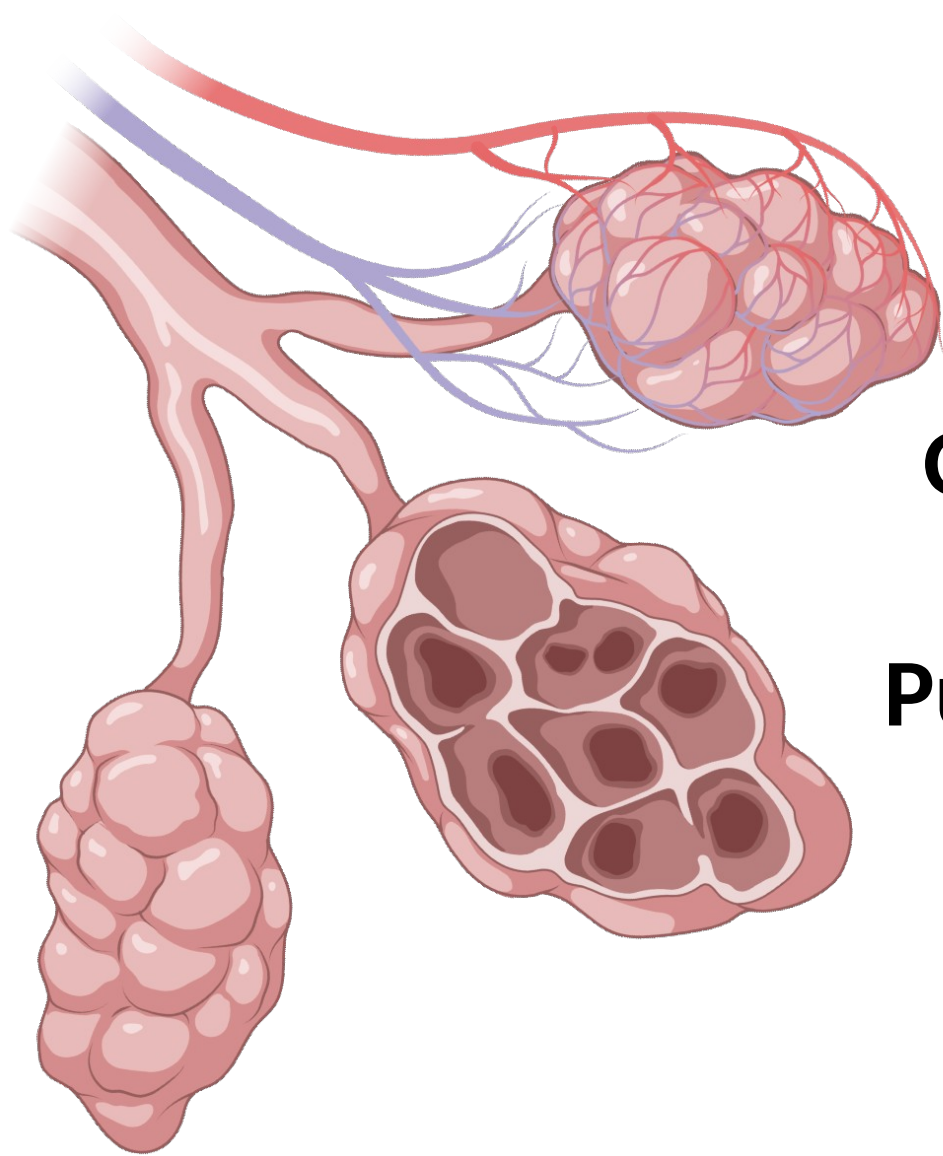
Classification

Management

Efficacy

Exacerbations

Cases



Outpatient Management of Chronic Obstructive Pulmonary Disease (COPD)

Diagnosis

Classification

Management

Efficacy

Exacerbations

Cases

Diagnosis

Criteria for diagnosis?

Initial evaluation

1. Risk Factors

Tobacco smoke
Other smoke, vapor, or dust exposure
Host factors (genetics, prematurity, frequent respiratory infections)

Tobacco use history
Exposure history
Consider α -1 antitrypsin

2. Symptoms

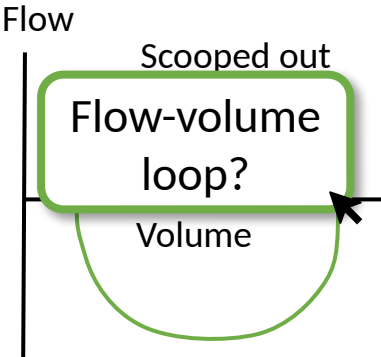
Dyspnea
Chronic cough
Sputum production
Recurrent LRTI

CXR
ABG (if $\text{SpO}_2 < 92\%$)
CT chest (FEV1 <45%, severe symptoms, lung cancer screen)

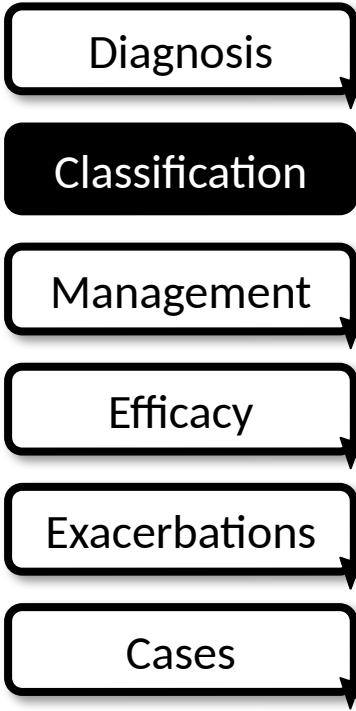
3. Spirometry

Post- bronchodilator
FEV1/FVC ratio <0.7 is mandatory for diagnosis!

Other PFT findings?



	% Ref (pre)	Post Meas	% Chg (post)
FEV1	30		7
FVC	75		8
FEV1/ FVC		0.47	
$\pm \uparrow$ TLC, $\uparrow$ RV, $\downarrow$ DLCO			



Classification

Scores of symptom severity

Modified Medication

mMRC

Borg Council Dyspnea Scale

COPD

CAT

Test

Assesses health status in COPD

Each scored 1-5

- Cough
- Phlegm
- Chest tightness
- Breathlessness

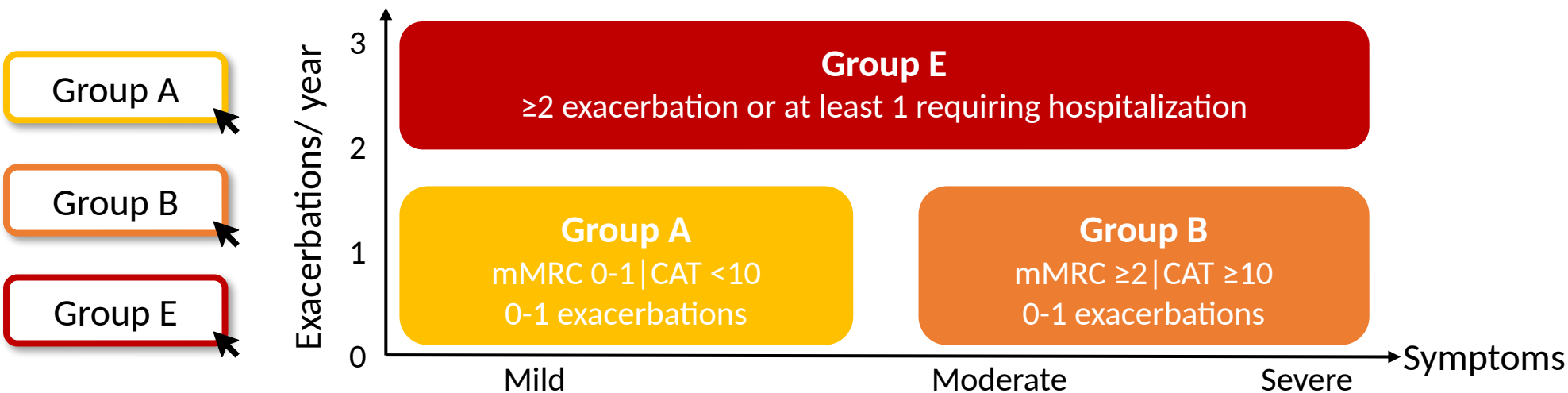
- Activities
- Confidence
- Sleep
- Energy

Score <10

Score >10

Score	Symptom severity (Dyspnea)
0	Strenuous exercise
1	Hurrying, walking up slight hill
2	Walking
3	<100 y or few min
4	With ADLs

COPD classification



Diagnosis



Classification



Management

Efficacy



Exacerbations



Cases



Management

Group A



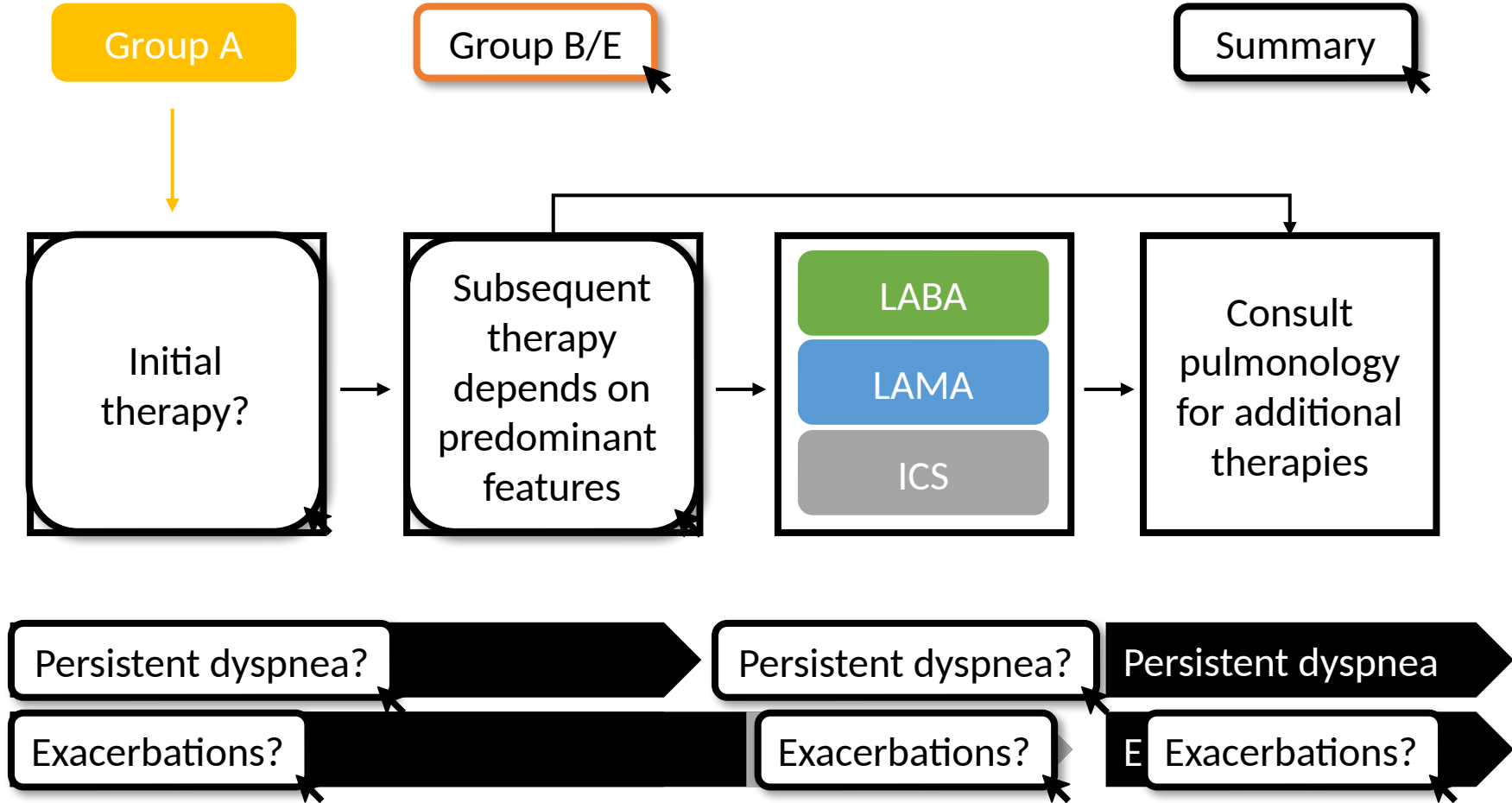
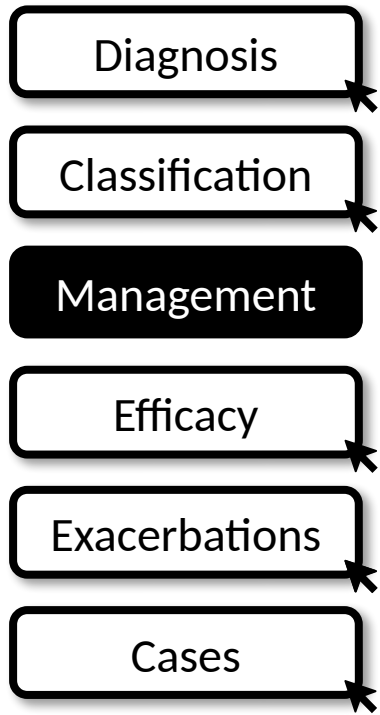
Group B/E

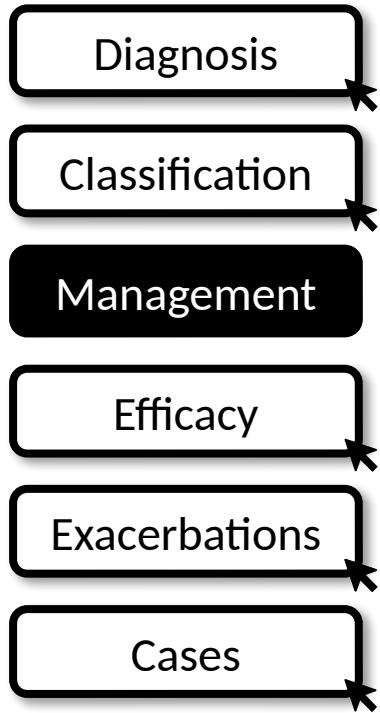


Summary

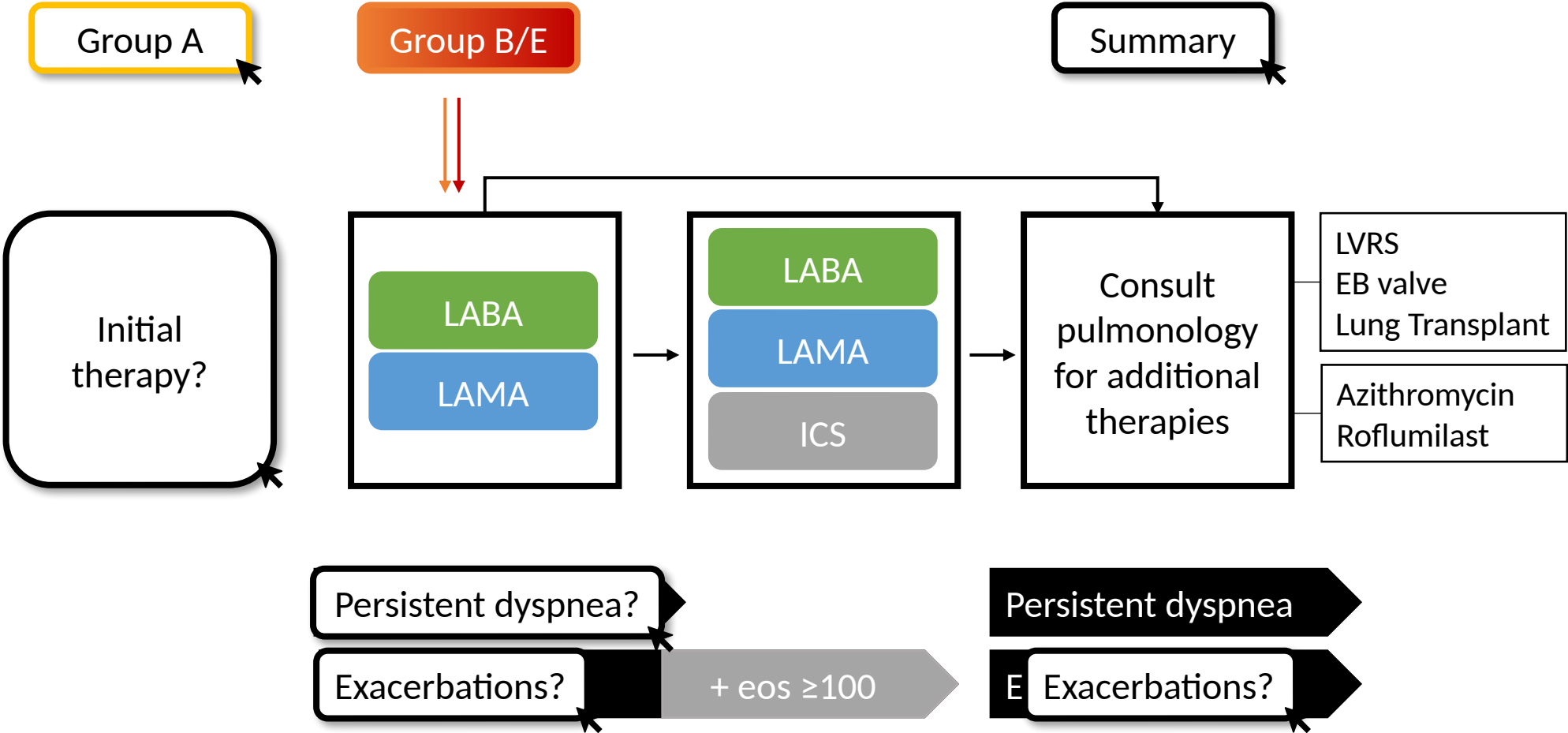


Management



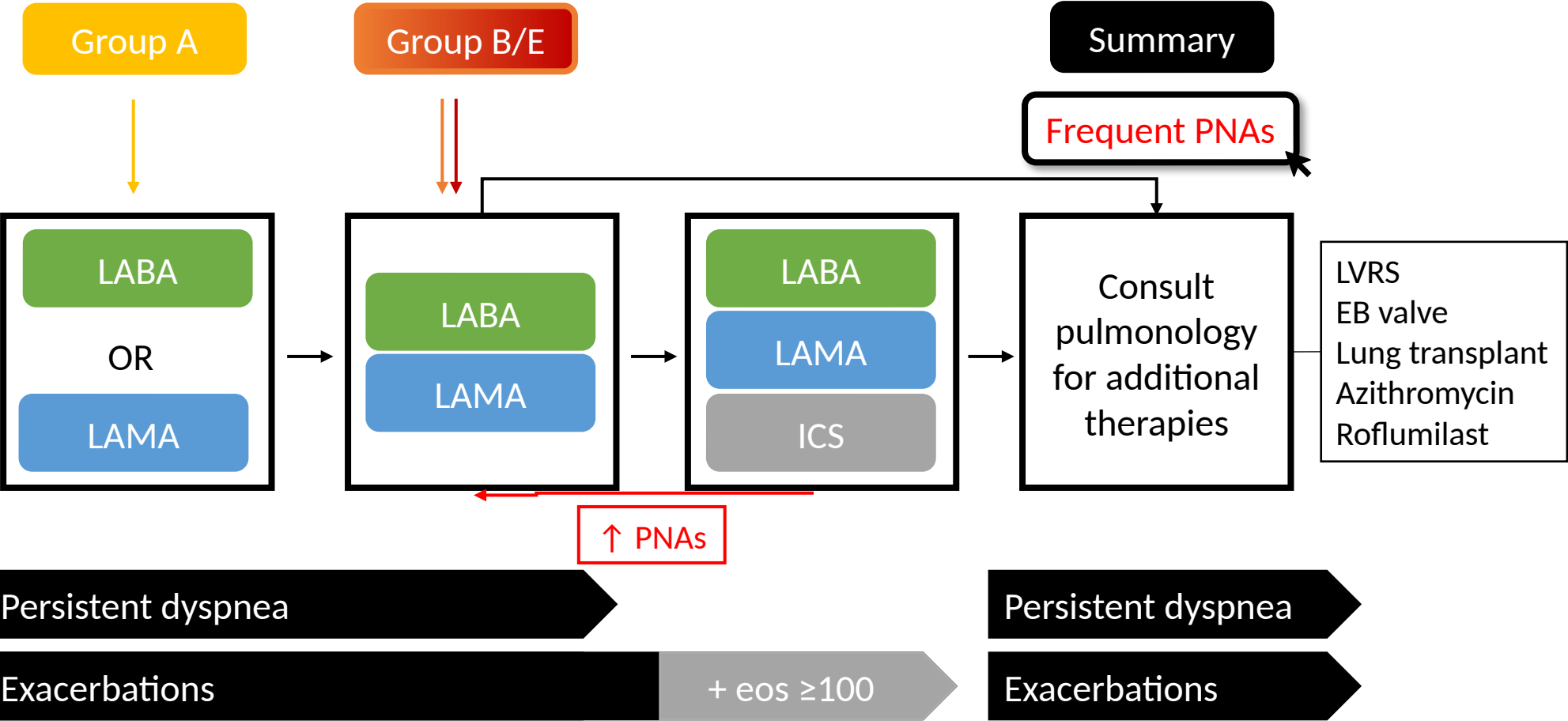


Management



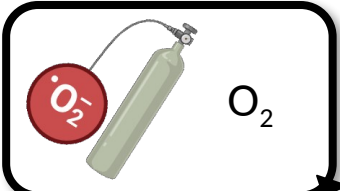
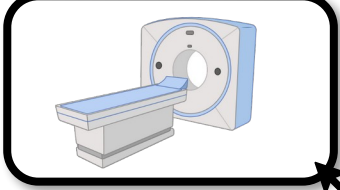
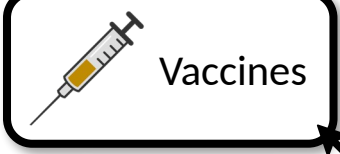
- Diagnosis
- Classification
- Management
- Efficacy
- Exacerbations
- Cases

Management



Diagnosis
Classification
Management
Efficacy
Exacerbations
Cases

Efficacy of nonpharmacologic therapies

	Reduce exacerbations	Reduce symptoms	Improve Function	Slow Progression	Prevent Death
 Smoking cessation	+	+	+	+	+
 O ₂		+	+	+	+ PaO ₂ <55 or PaO ₂ <60 +RHF
	+	+	+		+
					+
 Vaccines	+				? For severe COPD

Acute Exacerbations

Diagnosis

Classification

Management

Efficacy

Exacerbations

Cases

Acute
exacerbation?

Cardinal symptoms:
↑ dyspnea
↑ sputum volume
↑ sputum purulence

Rule out
other causes

±CXR
±D-dimer
±Flu swab
±BNP, ECG, trop

Hospitalization?

- **Severe exacerbation:** resting dyspnea, ↑ O2 requirement, confusion
- Severe comorbidities
- Insufficient home support

Steroids?

If new symptoms **limit daily activities.**
↓ rate of relapse and symptoms.

Dose and
duration?

Prednisone 40 mg
x 5 d.

Antibiotics?

If **mod-severe exacerbation** +
≥2 symptoms (including sputum
purulence)

High risk for poor
outcome or
Pseudomonas?

+

Anti-
pseudomonal
agent x 5 d

-

Azithromycin/
doxycycline

Bronchodilators?

SABA ± SAMA
MDIs as effective as nebulizers when used appropriately

Diagnosis

Classification

Management

Efficacy

Exacerbations

Cases

1-2

3

Case 1

60-year-old with 60 pack-year of smoking presents with progressive shortness of breath and cough. They report breathlessness when walking 100ft on flat ground. No history of hospitalizations for these symptoms.

What additional work-up is needed to evaluate his respiratory symptoms?

Labs: CBC (absolute eos = 250), BNP
CXR – hyperinflation, no other notable findings
Spirometry – post bronchodilator FEV1 60% and FEV1/FVC = 55%

Diagnosis?

COPD Group B

Next step in management?

LABA + LAMA + PRN SABA or SAMA
Smoking cessation
Vaccinations – annual influenza, PPSV23
Pulmonary rehabilitation therapy

Inhaler selection?

One or two short acting: SABA and/or SAMA PRN
AND a long-acting bronchodilator: LAMA or LABA Scheduled

Case 2

60-year-old (former smoker) with COPD (FEV1 60%) presents with 1 week of ↑ dyspnea and productive cough preventing limiting strenuous activity, but able to perform activities of daily living. Denies new allergen exposure or sick contacts. Has had 1 prior exacerbation. No prior history of *Pseudomonas*.

Meds: Combivent (SABA/SAMA) PRN + salmeterol (LABA) daily.

Vitals: Afebrile, SpO2 90%, RR 20, diffuse wheezing.

What additional work-up is indicated?

Consider **CBC with differential, CXR**. Rule out alternative diagnoses

Where should this patient be treated?

Outpatient. They can perform ADLs.

Next step in management?

Steroids – prednisone 40mg PO daily x 5 days
Antibiotics – 5-day course of antibiotics (e.g., doxycycline, azithromycin) if sputum purulence.

Changes in long term therapy?

Add LAMA. Consider ICS if eos >100.

Diagnosis

Classification

Management

Efficacy

Exacerbations

Cases

1-2

3

Case 3

80-year-old former smoker with COPD (FEV1 35%) on 2 L home O₂ presents for gradually worsening dyspnea, now affecting some ADLs. No exacerbations within the last year. He has no history of colonization with *Pseudomonas*.

Meds: LABA-LAMA daily + PRN SABA

What additional work-up is needed?

Rule out other causes of worsening dyspnea. Consider CXR, BNP, CBC for anemia, etc.

Next step in management?

- **Observe inhaler technique**
- Ensure adherence to home O₂ therapy
- Consider pulmonary rehab
- Refer to Pulmonology for advanced therapies (LVRS, endobronchial valves).

....Pulmonology does not recommend additional therapies given age. The next month, they are admitted to the hospital for a COPD exacerbation and persistent dyspnea.

Next step in management?

Check peripheral eosinophils. **Consider adding ICS** if eos >100 with frequent exacerbations.
Consider **palliative care** for symptom management.

...They start a combination LABA-LAMA-ICS. Two months later, they are readmitted to the hospital with productive cough, found to have pneumonia and a COPD exacerbation.

Next step in acute COPD management?

- **Check an ABG** – patients with pH <7.35 and pCO₂>45 benefit from NIPPV.
- **Steroids** – prednisone 40 mg daily x 5 days.
- **Sputum culture** – consider for patients with sputum purulence to assess for bacterial colonization.
- **Antibiotics** – given recent exacerbation and severe COPD (FEV1 35%), treat with anti-PsA antibiotic (e.g., levofloxacin)

Next step in chronic COPD management?

- Consider de-escalating ICS therapy given pneumonia.
- Consider Pulm consult for advanced therapies (e.g., azithromycin).
- Palliative care consult and advanced planning.

2024–2025 GOLD Update — New Drug Classes & Refinements

Standing GOLD-2023 teaching (FEV1/FVC <0.7 dx, A/B/E groups, LABA-LAMA first-line, eos-guided ICS, 5-day steroid/abx for exacerbations) verified CURRENT in GOLD 2024/2025.

Triple therapy preferred: when ICS is indicated, LABA+LAMA+ICS is favored OVER LABA-ICS (better exacerbations, lung function, QoL).

Eosinophil thresholds (unchanged): initial ICS in Group E if eos ≥ 300 ; escalate to triple if exacerbating + eos ≥ 100 . De-escalation more likely to trigger exacerbations if eos ≥ 300 .

FIRST biologic for COPD — dupilumab (Dupixent): add to triple therapy if ongoing exacerbations + eos ≥ 300 + chronic bronchitis. FDA Sept 27, 2024 (BOREAS/NOTUS).

Novel inhaled class — ensifentrine (Ohtuvayre): dual PDE3/PDE4 inhibitor (bronchodilator + anti-inflammatory) by nebulizer; FDA June 26, 2024 (ENHANCE) — first new inhaled mechanism in ~20 yrs.

GOLD 2025 also emphasizes CVD/pulmonary-hypertension comorbidity screening and notes spirometry/imaging refinements.

Take-Home Points

In a patient with risk factors, post-bronchodilator FEV1/FVC <0.7 on spirometry (scooped flow-volume loop) is REQUIRED to diagnose COPD — you cannot diagnose COPD without PFTs.

Smoking cessation is the intervention with the clearest mortality benefit; supplemental O2 (PaO2 ≤55, or ≤59 + cor pulmonale/RHF) and vaccines also matter.

By 2023 GOLD groups: LABA+LAMA is first-line for most symptomatic patients (Groups B and E); add ICS mainly for exacerbators with eos ≥100 (initial ICS in Group E if eos ≥300). De-escalate ICS if recurrent pneumonia.

Treat outpatient exacerbations with short-acting bronchodilators, prednisone 40 mg ×5 d, and antibiotics ×5 d (per local resistance) when ≥2 cardinal symptoms incl. sputum purulence.

2024–25: dupilumab (first biologic, eos ≥300 + chronic bronchitis) and ensifentrine (PDE3/4) are new add-on options.

Attribution & Optimization Notes

Adapted from TeachIM.org — “COPD – Outpatient Management” (McLeod, Guinn, Brett, Fainstad; Section Ed. Steinbach; Univ. of Colorado).
Published Sept 2021, last reviewed May 2023.

Source URL: https://teachim.org/teaching_material/copd-outpatient/ | Deck: copd-2023.pptx (embed-verified)

Licensed under Creative Commons Attribution-NonCommercial 4.0 (CC BY-NC 4.0). This optimized package is likewise CC BY-NC 4.0 — attribution required, non-commercial use only.

Optimized June 2026: original interactive board preserved (all 13 slides, animations & BioRender graphics intact); no projected-text factual/spelling errors found → append-only edit (this 3-slide addendum). Speaker-notes typos left unchanged (non-projected).

Currency verified vs GOLD 2024 & GOLD 2025 reports and FDA approvals (ensifentrine Jun 2024; dupilumab Sept 2024).

Companion page-linked handout COPD-teacher-guide-2023.docx was NOT present in Source Files — flagged for upload. Educational use only — clinician-reviewed before teaching.